

# Dataset Details

## A. Domain Context

- 1. Peer-to-peer used-goods marketplaces
- 2. One-shot, short (~12-turn) text negotiations
- 3. Primary task: price bargaining of smartphones/electronics.

Primary Biases:-

- 1. Anchoring - first numbers strongly shape final prices

## B. Five Decision Factors

1 --- Customer Vulnerability (Language-Based) : Detected through budget stress, urgency, negative sentiment, and anxiety markers.

Threshold:  $\geq 0.6 \rightarrow$  Mandatory mitigation ( $\geq 1$  FCA vulnerability driver)

## 2 --- Customer Sophistication (Language + Behavior)

| Indicator            | Signals                                                         |
|----------------------|-----------------------------------------------------------------|
| Question Quality     | Specific comparative or technical questions show sophistication |
| Language Complexity  | Diverse vocabulary and conditional phrasing indicate expertise  |
| Market Awareness     | References to comparable prices and alternatives show research  |
| First Offer Strategy | Lower first offers relative to asking price show anchor control |

Threshold:  $\geq 0.6 \rightarrow$  Sophisticated enough for ethical leverage

3 --- Bias Strength (Numerical Anchoring Index)

Anchoring Strength =  $1 - (\text{Actual Adjustment} / \text{Rational Adjustment})$

Rule: Strong anchoring that harms the customer  $\rightarrow$  mitigation

- Asking Price (P) = Price listed by seller
- Anchor (A) = First price mentioned by customer
- Fair Value (F) = Estimated market value (typically 75% P)
- Current Offer (O) = Customer's latest offer
- Rational Adjustment =  $|F - A|$
- Actual Adjustment =  $|O - A|$

4 --- Harm Direction (Numerical Fairness Check)

| Customer Offer vs Fair Value | Effect      | Decision            |
|------------------------------|-------------|---------------------|
| > Fair Value                 | overpayment | Mitigate            |
| $\leq$ Fair Value            | benefit     | Customer advantaged |

Tolerance ( $\pm 10\%$ ) Justification: Baye et al. (2006) found online retail price dispersion SD = 10-15%; used goods = 15-25%. The  $\pm 10\%$  threshold captures normal market friction.

5 ---Vocal Stress & Emotional Instability Index (VSEI): for prosodic analysis

- \*\*VSEI Score Validation\*\*:**
- Calculate VSEI from prosodic features (as per your paper)
  - Compare against human-rated stress levels
  - **\*\*Correlation metric\*\*:** Pearson correlation between VSEI and human ratings

## **E. Dataset Specification (Dialogues)**

Scenario Distribution balanced across:-

1. Vulnerability (low/high)
2. Sophistication (low/high)
3. Anchor direction (harms/benefits customer)

Persona Requirements:-

1. Buyers: college student, urgent replacement seeker, first-time buyer, enthusiast, bargain hunter, etc.
2. Sellers: casual seller, upgrader, urgent mover, reseller

Dialogue Requirements:-

1.  $\geq 3$  customer price offers
2.  $\geq 2$  questions
3. Vulnerability signals where applicable
4. Market comparison signals for sophisticated personas
5. Customer has the bias/vulnerability/sophistication/harm direction/VSEI not vice versa.